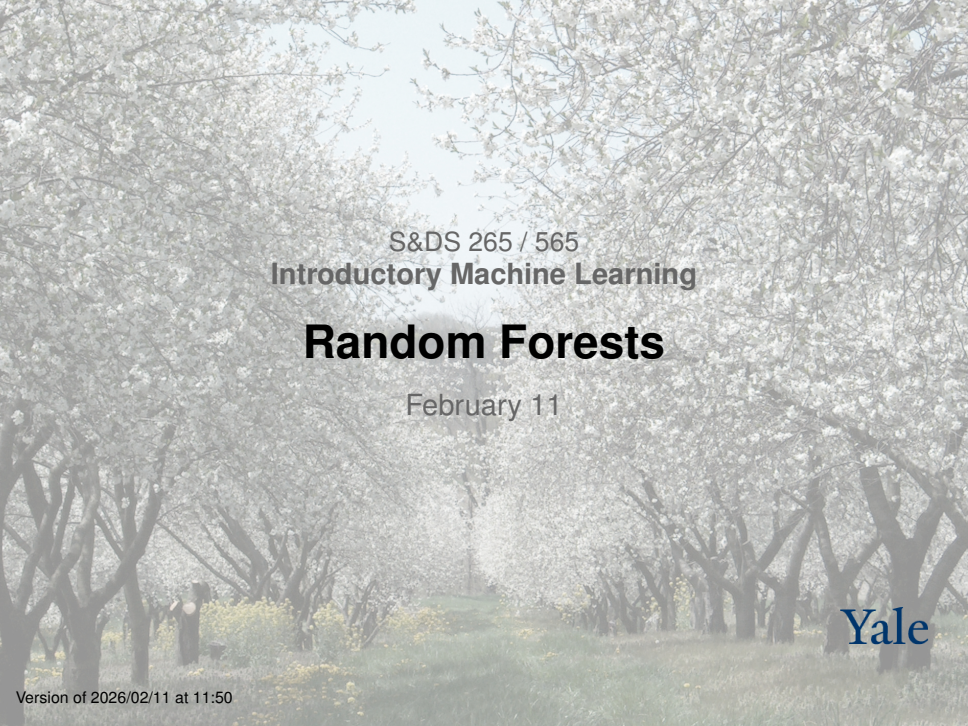




S&DS 265 / 565
Introductory Machine Learning

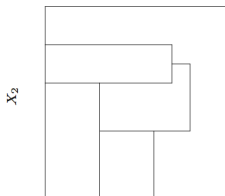
Random Forests

February 11

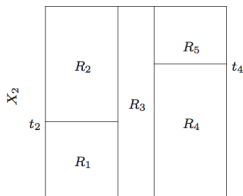
Yale

Friendly reminders

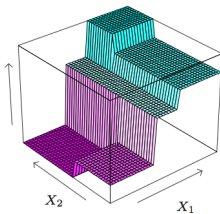
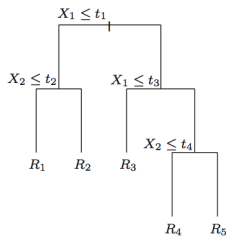
- Assn 2 due next week
- Quiz 2 live after class today
- Midterm March 4, in class
- Practice midterms (and solutions) posted on Canvas
- Review sessions will be organized



X_1



X_1



Trees vs. other methods

Decision trees are similar in spirit to k -nearest neighbors.

Trees vs. other methods

Decision trees are similar in spirit to k -nearest neighbors.

- Both produce simple predictions (averages/majority vote) based on “neighborhoods” in the predictor space.

Trees vs. other methods

Decision trees are similar in spirit to k -nearest neighbors.

- Both produce simple predictions (averages/majority vote) based on “neighborhoods” in the predictor space.
- Neighborhoods chosen very differently

Trees vs. other methods

Recall that linear regression fits models of the form

$$f(X) = \beta_0 + \sum_{j=1}^p X_j \beta_j$$

Trees vs. other methods

Recall that linear regression fits models of the form

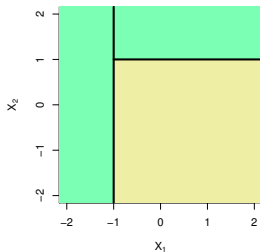
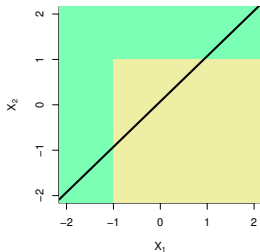
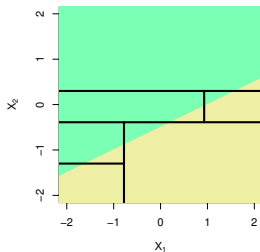
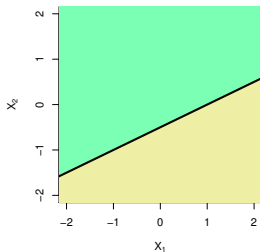
$$f(X) = \beta_0 + \sum_{j=1}^p X_j \beta_j$$

Regression trees are like fitting linear regression models with a bunch of indicators

$$f(X) = \sum_{j=1}^J \beta_j \mathbb{1} \{X \in R_j\}$$

Trees vs. other methods

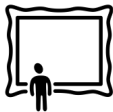
Are trees always better than linear methods?



Summary so far

- Trees give interpretable, nonlinear prediction rules
- Deep trees have low bias, high variance
- Shallow trees have high bias, low variance
- Deep trees are pruned back using cross-validation to find best bias/variance tradeoff.

Random Forests



Suppose that $Y_1, Y_2, \dots, Y_n \sim N(\theta, \sigma^2)$.

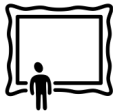
Let $\bar{Y}_n$ be the average $\bar{Y}_n = \frac{1}{n}(Y_1 + Y_2 + \dots + Y_n)$. Then

$$\mathbb{E}(\bar{Y}_n) = \theta$$

If Y_i are uncorrelated then

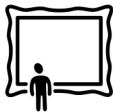
$$\text{Var}(\bar{Y}_n) = \frac{\sigma^2}{n}$$

Random Forests



- Grow many trees and average their predictions
- Trees are grown deep, to have low bias, but high variance
- To “decorrelate” the trees and reduce variance, introduce randomness

Random Forests



- Grow many trees and average their predictions
- Trees are grown deep, to have low bias, but high variance
- To “decorrelate” the trees and reduce variance, each tree is
 - ▶ grown on a bootstrap sample of the data
 - ▶ grown with random subsets of the predictors at each split
- Tree growing can be done in parallel

Leo Breiman—"Keep it simple"



Random Forests Algorithm

- ① For $b = 1$ to B :
 - (a) Draw a bootstrap sample Z^* of size n from the training data
 - (b) Grow a random-forest tree T_b to the bootstrapped data, recursively repeating following steps, until minimum node size reached:
 - i. Select m variables at random from the p variables
 - ii. Pick the best variable/split-point among the m
 - iii. Split the node into two children nodes
- ② Output the ensemble of trees $\{T_b\}_{b=1}^B$.

Random Forests Algorithm

To make a prediction at a new point x :

Regression: Average $\hat{f}_{rf}^B(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$

Classification: Majority vote of the individual trees

Mining and Forecasting NFL Drive Statistics via the ESPN API using Random Forest Classification to Predict Game Outcomes



Andy J. Paladino

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23 min read · Nov 21, 2024



3



2



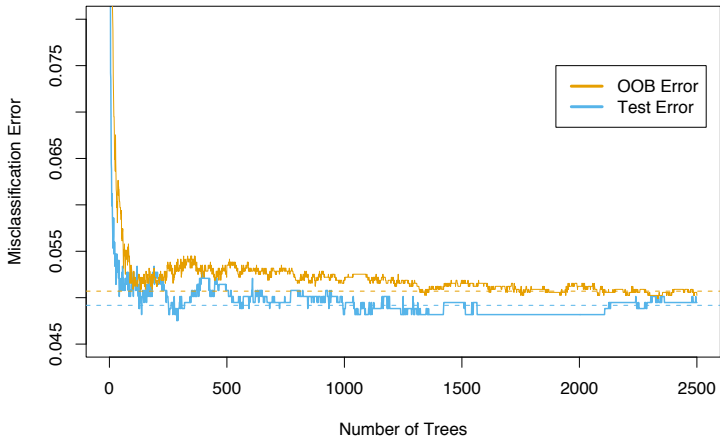
Out of bag (OOB) prediction

A bootstrap sample, or “bag” of data, is a set of n data points sampled with replacement from the original set of n data points. Will contain repetitions.

Each tree is grown on such a sample, which contains about $\frac{2}{3}$ of the original data (with repetitions). The remaining $\frac{1}{3}$ can be used as validation data

- For each observation $z_i = (x_i, y_i)$, construct its random forest predictor by averaging only those trees corresponding to bootstrap samples in which z_i did not appear
- Thus, cross-validation can be performed “along the way”

Out of bag (OOB) prediction



More detail: Bootstrap samples

- We have n data points
- Randomly sample exactly n of them *with replacement*
- Contains about $\frac{2}{3}$ of the data, with duplicates
- This is a “bag” of n data points

Chance a particular example does not appear in sample:

$$\left(1 - \frac{1}{n}\right)^n \rightarrow \frac{1}{e} \approx 0.37$$

More detail: Out-of-Bag error estimation

Each bagged tree uses about $2/3$ of all observations (with repeats).

The remaining data—*out-of-bag* (OOB) observations—can be put to good use.

More detail: Out-of-Bag error estimation

- Consider bags $\mathcal{B}_i \subset \{1, 2, \dots, B\}$ where (x_i, y_i) does *not* appear
- We average over those trees to form a prediction:

$$\hat{y}_i = \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} T_b(x_i)$$

- OOB error estimate is then $\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$

Performance on email spam task

Single tree: 8.7%

Random forest: 5.1%

(standard error of the estimates is $\approx 0.6\%$)

Variable importance

Random forests improve upon the predictive ability of trees, but this sacrifices the interpretability of the single tree model

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Random forests improve upon the predictive ability of trees, but this sacrifices the interpretability of the single tree model

A good tool for interpreting a forest is **variable importance**

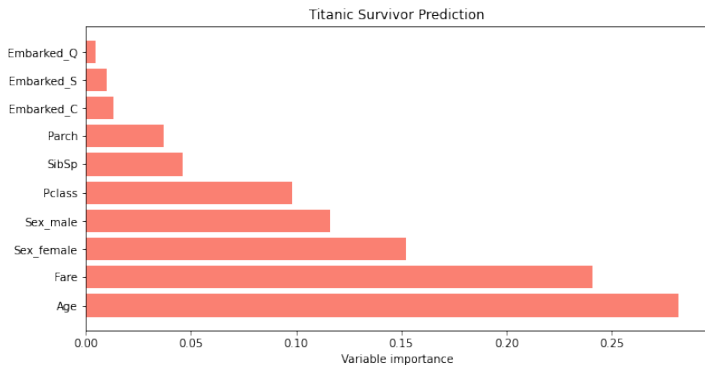
Variable importance

Random forests improve upon the predictive ability of trees, but this sacrifices the interpretability of the single tree model

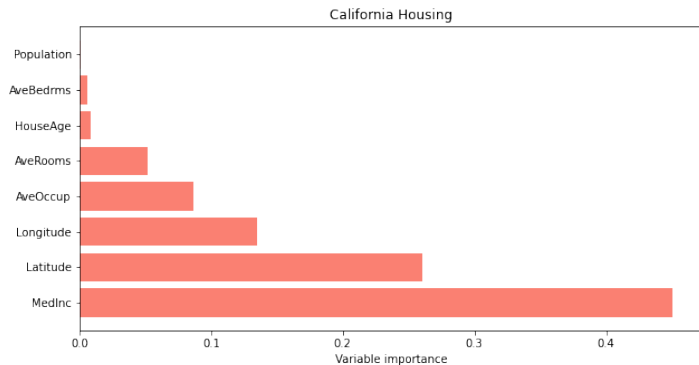
A good tool for interpreting a forest is **variable importance**

Variable importance is the amount that the RSS (or other loss) is reduced due to splits over a given predictor, averaged over all trees in the forest

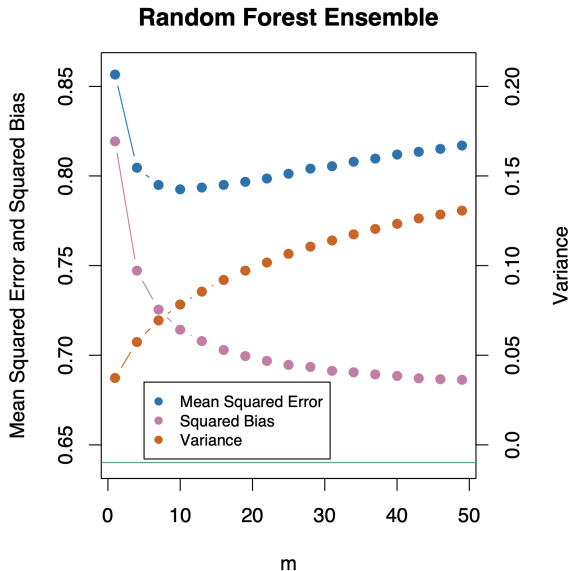
Variable importance



Variable importance



Random forest MSE



Understanding the bias-variance tradeoff for random forests

- Each tree has low (squared) bias, because it is deep
- The bootstrap sample and random subset of questions allowed at a given split result in “diversity” among of the trees
- This diversity translates to decorrelation, and tends to reduce the variance
- As we increase the number m in the random subset of questions allowed at each split, the complexity increases—variance goes up, squared bias goes down

Please step through the notebook

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

```
In [2]: titanic_train = pd.read_csv('https://raw.githubusercontent.com/minsuk-heo/kaggle-titanic/master/input/train.csv')
titanic_test = pd.read_csv('https://raw.githubusercontent.com/minsuk-heo/kaggle-titanic/master/input/test.csv')
titanic_train
```

Out[2]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

891 rows x 12 columns

Summary: Trees and Forests

- Trees give interpretable, nonlinear prediction rules
- Deep trees have low bias, high variance
- Random forests are a way of combining trees
- Want: Different trees should capture different aspects of the data
- How: Grow each tree on a random (bootstrap) sample of the data and choosing from a random set of questions at each split